

Online Retail Customer Analytics

A UK-based online gift retailer, December 2010 – December 2011:
where the revenue comes from, who the customers are, and what to do next.

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Executive Summary

A year of invoice data — **541,909 line items, 25,900 invoices, £9.75 million in net revenue** — reveals a UK gift retailer that operates, in practice, as a **domestic wholesaler with a thin export layer**. The UK accounts for **91.4% of transactions** and **84% of revenue**; no other country clears 2% of transaction volume. Demand is highly seasonal: **November delivers roughly 15% of annual revenue**, while February is the trough. Value is concentrated — one customer spent £279,489 and the top ten account for 17.3% of completed-sales revenue. Operations are healthy (a 1.71% cancellation rate) but leak in one place: **24.9% of line items carry no CustomerID**, most of them UK transactions, leaving a quarter of domestic sales invisible to retention marketing. RFM segmentation sharpens the retention picture: 557 Champions to protect, 618 At-Risk customers to win back, and 25 high-value "Cannot Lose Them" customers averaging £2,123 in spend who have not ordered in ~231 days — the single highest-value win-back pool in the base.

The Business in Numbers

Metric	Value
Invoice line items	541,909
Invoices	25,900
Net revenue (cancellations included as negatives)	£9,747,748
Customers with completed sales	4,339
UK share of transactions	91.4%
UK share of revenue	84.0%
Cancellation rate (all line items)	1.71%

Key Findings

1. A domestic business with a thin export layer

The UK is 91.4% of transactions; only Germany (1.75%), France (1.58%) and EIRE (1.51%) individually clear 1%. Revenue concentration mirrors volume: the UK contributes £8,187,806, followed by the Netherlands (£284,662), EIRE (£263,277), Germany (£221,698), France (£197,404) and Australia (£137,077) — six countries above £130,000. Growth strategy should assume a domestic core; export is diversification, not the engine.

2. Wholesale rhythm: seasonal, weekday, midday

The buying pattern is wholesale, not consumer. **Thursday (103,857 transactions)** is the busiest day; weekends are quiet. The annual peak lands in **November (84,711)** — pre-Christmas stocking — with **February (27,707)** as the trough and December 2011 a partial month (25,525, dataset ends 9 December). The busiest hour is 12:00 GMT (78,709 transactions); the operating-day minimum is **20:00 GMT** (871), the natural window for site maintenance.

3. Value concentrated in a few hands

Customer 14646 alone generated £279,489 in net revenue, and the top ten customers account for 17.3% of completed-sales revenue (£8,911,408). The single biggest "product" by revenue is DOTCOM POSTAGE at £206,245.48 — delivery fees are a revenue line in their own right, worth pricing attention.

4. Healthy operations, one data leak

The overall cancellation rate is a negligible 1.71%; among major markets the highest is Australia at 5.88% (74 of 1,259), with the UK at 1.59% and France at 1.74%. The real leak is attribution: 24.9% of line items have no CustomerID, and 133,600 of those are UK transactions. A quarter of domestic sales cannot be tied to a customer for retention marketing — the single biggest hole in the retention engine.

5. RFM: a clear retention map

Segmenting the 4,339 customers with completed purchases produces a directly actionable map:

Segment	Customers	What it means
Need Attention	858	Largest pool; re-engagement campaigns
At Risk	618	~3 orders historically, ~149 days since last purchase
Potential Loyalists	592	Recent buyers to nurture upward
Loyal Customers	582	Steady core; protect and upsell
Champions	557	Best customers; reward and retain
Lost	556	Win back selectively
Hibernating	509	Long-dormant; low-cost reactivation
New Customers	42	Onboarding journeys
Cannot Lose Them	25	Avg £2,123 spend, ~231 days since order — personal outreach

Cohort behavior is wholesale-typical: the December 2010 cohort of 885 customers holds roughly 35–40% monthly activity, with a 50.3% spike in November — seasonal buyers return for the peak, then drift.

Recommendations

1. Win back the 25 "Cannot Lose Them" customers personally. A call or tailored offer costs little against £2,123 average spend each, and they have been silent for ~231 days. This is the highest-value win-back pool in the customer base.

2. Fix UK guest checkout. 133,600 unattributable UK transactions is the single biggest leak in the retention engine. Even a modest account-creation incentive — a small discount or faster reordering — feeds RFM segmentation directly and converts anonymous volume into targetable customers.

3. Move acquisition spend to September–November; treat February as the clearance window. Demand concentrates before Christmas, so marketing budget should lead the peak

rather than chase it. Schedule site maintenance at 20:00 GMT, the operating-day activity minimum.

Methodology & Provenance

All figures in this report are **Python** outputs computed from the UCI Online Retail dataset (541,909 rows, December 2010 – December 2011). Revenue is **net**: cancellations are included as negative line items (£9,747,748 total). Customer counts (4,339), RFM segmentation, cohort analysis and the top-10 concentration figure use **completed sales only** (33 customer IDs appear solely in cancelled invoices and are excluded from customer analysis). Every figure was traced to an executed output; no numbers were introduced from memory.